



Shahid Beheshti University

Department of Computer Science
Advanced Data Mining / Deep Learning

Mini Project 2

AE-1: ECG Anomaly Detection

Autoencoder Reconstruction on ECG5000

Project Information

Architecture family:	Autoencoder
Dataset:	ECG5000
Project value:	20 points
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Notebook Link:	Kaggle Notebook
Project Link:	GitHub Repository

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Data Mining Course

1 Problem and Data

This project studies binary anomaly detection on ECG5000, a collection of 5,000 pre-segmented heartbeat sequences, each represented by 140 ordered samples. The original dataset contains five classes. Following the notebook protocol, class 1 is treated as normal and classes 2–5 are grouped as abnormal for detection, while the original labels are retained for subtype-level error analysis.

The executed notebook used the standard original TRAIN split (500 beats) and TEST split (4,500 beats). Across the complete dataset there were 2,919 normal beats and 2,081 abnormal beats: 1,767 from class 2, 96 from class 3, 194 from class 4, and 24 from class 5. Only the original TRAIN partition was divided, using a stratified 70/30 split, into 350 training and 150 validation examples. This produced 204 normal and 146 abnormal training examples, 88 normal and 62 abnormal validation examples, and 2,627 normal and 1,873 abnormal test examples. The original TEST partition was untouched during fitting and threshold selection.

Preprocessing was also restricted to the training information allowed by the anomaly-detection design. A per-timestep standard scaler was fitted only on the 204 normal training beats and then applied to all splits. The Autoencoder itself was trained only on these normal training beats. Normal validation beats were used for early stopping and for the fixed-percentile threshold, whereas the mixed labeled validation set was used only by the supervised threshold-calibration contribution. This separation avoids fitting the representation or preprocessing statistics to anomalies or to the final test set.

2 Method

2.1 Dense Autoencoder and anomaly score

The principal model was a compact dense Autoencoder with architecture

$$140 \rightarrow 64 \rightarrow 32 \rightarrow 8 \rightarrow 32 \rightarrow 64 \rightarrow 140.$$

Hidden layers used ReLU activations and the output layer was linear. The model contained 22,868 trainable parameters. It was optimized with Adam at learning rate 10^{-3} using mean-squared reconstruction error. Training used batch size 128 for at most 200 epochs; early stopping monitored normal-validation loss with patience 20 and restored the best weights. In the executed run, training reached the 200-epoch limit, with the best validation loss occurring at epoch 199.

For a heartbeat $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$, the anomaly score was the per-sample reconstruction MSE,

$$e(\mathbf{x}) = \frac{1}{140} \sum_{t=1}^{140} (x_t - \hat{x}_t)^2.$$

A sample was predicted anomalous when $e(\mathbf{x}) > \tau$. Thus the Autoencoder produces a continuous ranking score, while the threshold τ determines the final precision–recall operating point.

Because abnormal and normal examples are not equally represented and false negatives are important in anomaly detection, Precision, Recall, and F1 were treated as the primary threshold-dependent metrics. Accuracy was retained as a secondary summary. Average Precision and PR-AUC were used to assess the common reconstruction-error ranking independently of a particular threshold.

3 Original Contribution

3.1 Hypothesis and controlled threshold intervention

The notebook states the following hypothesis: *a validation-optimized reconstruction-error threshold will improve Recall and F1 relative to a fixed 95th-percentile threshold calibrated on normal validation beats, with only a limited Precision trade-off.*

The comparison was deliberately controlled. The trained Autoencoder and all reconstruction errors were held fixed; only the threshold-selection rule changed. The baseline threshold was

$$\tau_{95} = Q_{0.95}(\{e_i : y_i = 0, i \in \text{validation}\}) = 0.700382.$$

The proposed threshold was selected from the validation precision–recall curve by maximizing validation F1 on the mixed labeled validation set, yielding

$$\tau^* = 1.350430.$$

The optimized rule is therefore a supervised threshold-calibration method because it uses labeled validation anomalies, whereas the baseline uses only normal validation errors.

On validation, the optimized threshold increased F1 from 0.9531 to 0.9756. The final untouched test set showed the opposite overall outcome: Precision increased, but Recall and F1 fell substantially. This negative result is central to the contribution because it demonstrates that an apparently better validation operating point did not generalize.

Table 1: Final test metrics for the two threshold-selection rules.

Method	Threshold	Precision	Recall	F1	Accuracy
95th-percentile baseline	0.7004	0.9439	0.9797	0.9615	0.9673
Validation-optimized	1.3504	0.9742	0.8655	0.9166	0.9344

Relative to the baseline, the optimized threshold changed Precision by +0.0302, Recall by −0.1143, and F1 by −0.0449. A paired bootstrap with 2,000 resamples gave 95% confidence intervals of [0.0227, 0.0384] for Δ Precision, [−0.1289, −0.0996] for Δ Recall, and [−0.0539, −0.0359] for Δ F1. The intervals support a genuine Precision gain but also a genuine Recall and F1 loss. The stated hypothesis was therefore **not supported** on the final test set.

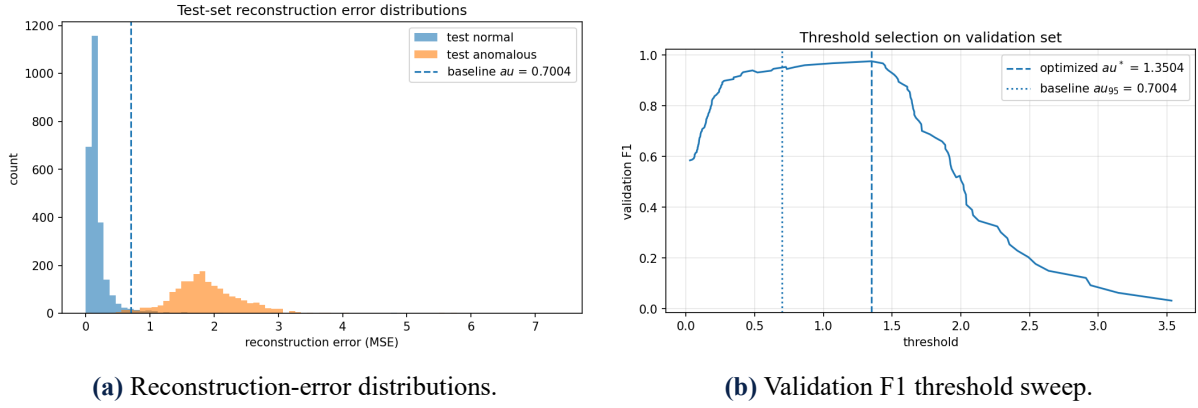


Figure 1: The anomaly score separates the two binary groups strongly, but validation F1 maximization selects a higher and more conservative test operating point than the normal-only percentile rule.

4 Experiments

The notebook evaluated the threshold contribution and then added targeted experiments to test score separation, uncertainty, failure modes, the two required discussion questions, robustness, and the handbook stretch goal.

4.1 Score separation and statistical analysis

On the test set, normal reconstruction errors had mean 0.229973 and median 0.128479, whereas abnormal errors had mean 2.041810 and median 1.825507. A one-sided Mann–Whitney U test asked whether abnormal beats tend to receive larger errors than normal beats. It returned $U = 4,852,239$ with a p-value below floating-point resolution in the notebook. The common-language effect size was

$$P(e_{\text{abnormal}} > e_{\text{normal}}) = 0.986153.$$

This establishes strong ranking separation, but it does not select the optimal decision threshold. Consistently, the common score achieved Average Precision 0.9515 and PR-AUC 0.9514.

The paired bootstrap analysis was appropriate because both threshold rules were evaluated on exactly the same test heartbeats. Its confidence intervals, reported above, quantify uncertainty in the change caused solely by the thresholding rule.

4.2 Failure analysis

The threshold trade-off is clear in the confusion matrices. The baseline produced 109 false positives and 38 false negatives. Raising the threshold to the validation-optimized value reduced false positives to 43 but increased false negatives to 252. Hence the Precision gain was purchased by a large loss in sensitivity.

The original ECG5000 labels reveal that this loss is not uniform across anomaly types. Under the validation-optimized threshold, class 4 was the hardest subtype proportionally: 76 of 175 class-4 test beats were missed (miss rate 0.4343). Class 5 also had a high miss rate of

0.3636, although only 22 class-5 examples were present. Class 2 contributed the largest absolute number of misses, 153 of 1,590, but its miss rate was much lower at 0.0962; class 3 had a miss rate of 0.1744. These results show why aggregate F1 alone is insufficient for understanding the detector’s clinical-style failure pattern.

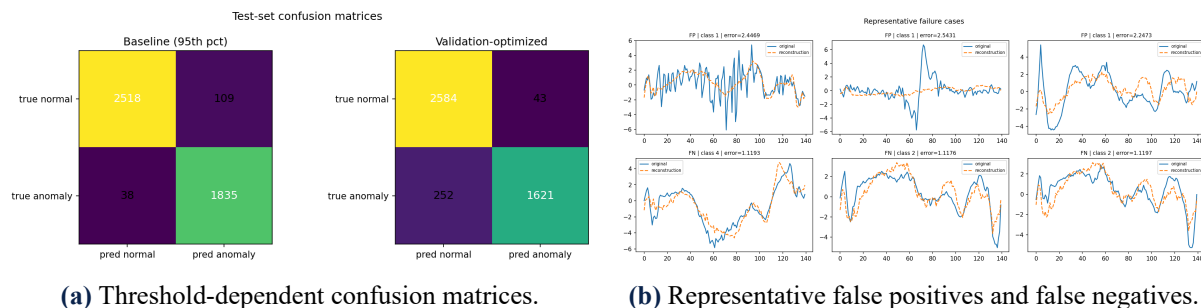


Figure 2: The optimized threshold suppresses false alarms but misses many more anomalies. Failure examples in the notebook were selected near the median error of each failure group rather than from extreme outliers.

4.3 Multi-seed robustness

The threshold comparison was repeated with seeds 7, 21, 42, 77, and 123 while holding the data split fixed. Only one of the five seeds produced a positive F1 change for the optimized threshold. Across seeds, the mean changes were

$$\Delta P = +0.0123 \pm 0.0221, \quad \Delta R = -0.0466 \pm 0.0689, \quad \Delta F1 = -0.0181 \pm 0.0244.$$

This reinforces the main conclusion: validation F1 maximization was not a stable improvement over the simpler percentile rule.

5 Discussion and Observation

5.1 Question 1: Reconstruction-capacity paradox

The handbook asks why a reconstruction model can become worse at anomaly detection when it becomes too capable. The required separation is not low reconstruction error by itself, but a persistent gap between normal and abnormal reconstruction errors. An overly expressive Autoencoder can approach an identity mapping and reconstruct abnormal morphology well, reducing that gap.

The notebook tested this idea by changing only the dense bottleneck size among 2, 8, and 32 dimensions while applying the same normal-only training and validation-optimized threshold procedure. To assess capacity, it reported both test F1 and the ratio of mean abnormal to mean normal reconstruction error.

Table 2: Bottleneck-capacity experiment. Error contrast is mean abnormal reconstruction error divided by mean normal reconstruction error.

Bottleneck	Error contrast	Precision	Recall	F1
2	11.1294	0.9685	0.9525	0.9604
8	8.8785	0.9742	0.8655	0.9166
32	7.2171	0.9323	0.9851	0.9579

The smallest bottleneck produced both the strongest error contrast and the highest F1. As the bottleneck widened, the error-contrast ratio decreased monotonically from 11.1294 to 8.8785 and then 7.2171. This pattern is consistent with the reconstruction-paradox mechanism: greater latent capacity made abnormal and normal reconstruction behavior less separated. However, the interpretation must remain limited to this experiment because only bottleneck width, not every component of model capacity, was varied. A practical sensitivity-preserving design would therefore favor compact bottlenecks, regularization or denoising, early stopping, and training restricted to representative normal examples rather than maximizing reconstruction capacity indiscriminately.

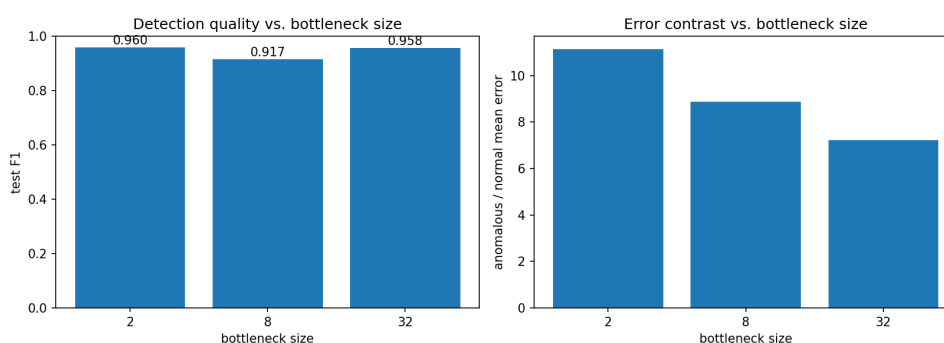


Figure 3: Detection F1 and reconstruction-error contrast across bottleneck sizes.

5.2 Question 2: What self-attention reveals, and whether it helps

Self-attention allows each temporal position to aggregate information from distant positions instead of relying only on local convolutional neighborhoods. In an ECG reconstruction model, its weights can therefore indicate which time regions are repeatedly used as context when the network reconstructs the heartbeat. Such weights are not causal explanations by themselves, so the notebook paired visualization with a controlled architectural ablation.

Two sequence Autoencoders used the same Conv1D reconstruction backbone and the same normal-only training protocol. The sole architectural intervention was a two-head self-attention block inserted between encoder and decoder in the attention model. The no-attention model had 5,345 parameters; the attention model had 9,633. Each model selected its threshold from the same validation procedure before evaluation on the untouched test set.

Table 3: Controlled self-attention ablation.

Model	Precision	Recall	F1	Error contrast	Parameters
No attention	0.7726	0.8398	0.8048	2.2669	5,345
Self-attention	0.9030	0.9744	0.9373	9.6741	9,633

Self-attention increased F1 by 0.1325, Recall by 0.1345, and the error-contrast ratio by 7.4072. Therefore, within this controlled setup, attention did more than merely add parameters: it was associated with substantially better anomaly discrimination under the same training protocol.

The temporal reconstruction-error profile found the largest abnormal–normal error gaps at time steps [5, 4, 16, 6, 136, 18, 15, 133]. Independently, the learned attention profile placed its highest average key importance at [5, 6, 7, 4, 123, 124, 122, 125, 126, 121]. The overlap around steps 4–7 indicates that the attention module repeatedly revisited a region that was also strongly discriminative in reconstruction-error analysis. Its additional emphasis near the end of the sequence suggests use of long-range context from both early and late heartbeat regions.

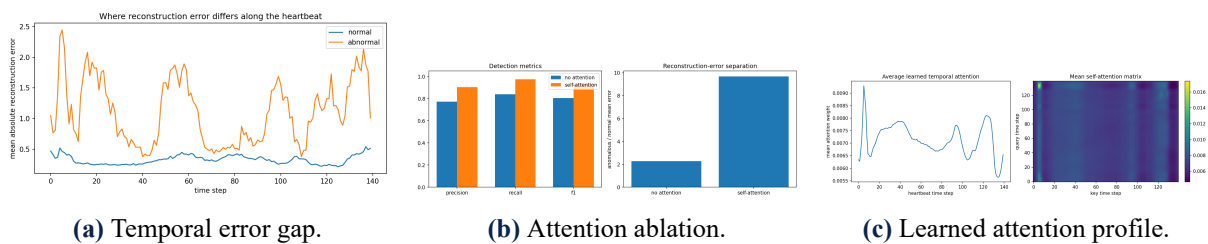


Figure 4: Project-specific evidence for the attention discussion: error localization, controlled performance comparison, and the learned temporal attention distribution.

6 Stretch Bonus: Isolation Forest Baseline

The AE-1 card in the handbook proposes comparison with a one-class SVM or Isolation Forest as a stretch bonus. The notebook completed this stretch experiment using Isolation Forest with 300 estimators. It was fitted only on normal training beats. To keep its decision protocol comparable with the fixed Autoencoder baseline, its anomaly-score threshold was the 95th percentile of normal validation scores.

Table 4: Stretch comparison with the Isolation Forest one-class baseline.

Method	Precision	Recall	F1	Accuracy
AE, 95th-percentile threshold	0.9439	0.9797	0.9615	0.9673
AE, validation-optimized threshold	0.9742	0.8655	0.9166	0.9344
Isolation Forest	0.7246	0.1335	0.2254	0.6182

Isolation Forest achieved Average Precision 0.8274 and was markedly weaker at the selected operating point, especially in Recall. This comparison supports the usefulness of learned

reconstruction features for this dataset, while also showing that performance depends on both scoring method and threshold calibration. The bonus result should not be interpreted as a universal superiority claim for Autoencoders because only one Isolation Forest configuration and one dataset were evaluated.

7 Limitations and Next Steps

The original TRAIN partition is small, leaving only 204 normal examples for Autoencoder fitting and 150 total validation examples for calibration. The main proposed threshold is supervised because it uses labeled validation anomalies, and its validation improvement did not generalize to the untouched test set. The multi-seed experiment also showed that its F1 effect was unstable.

The dense baseline does not encode local temporal structure explicitly, while the sequence attention experiment changes both parameter count and architecture; although its controlled ablation is informative, a parameter-matched attention control would isolate the source of the gain more tightly. The bottleneck experiment varies one dimension of capacity rather than total depth, regularization, or training duration. Temporal-error and failure analyses are post-hoc test-set analyses and were correctly not used for further tuning.

Future work should evaluate threshold objectives that explicitly constrain Recall, use repeated or cross-validated calibration rather than a single small validation split, and test temporal architectures with parameter-matched controls. Subtype-aware diagnostics are also important because class 4 and class 5 showed substantially larger miss rates than class 2. More broadly, the project demonstrates that strong anomaly-score separation does not guarantee that a validation-selected decision threshold will transfer reliably to an unseen test distribution.